

Objective

Study of Hardware and Software Requirements for Different Operating Systems (Linux, Unix, Windows XP, Windows 7/8).

Theory

An Operating System (OS) is system software that manages computer hardware and software resources and provides services to computer programs. Different operating systems have different hardware and software requirements based on their design, performance, and usage.

This experiment focuses on understanding the minimum and recommended requirements for popular operating systems.

Hardware and Software Requirements

Hardware requirements refer to the physical components needed to install and run an operating system properly. These include processor speed, RAM, hard disk space, display resolution, and peripheral devices.

Software requirements refer to the system-level software components needed for the installation and smooth functioning of an operating system.

Operating System 1: Linux

Introduction to Linux

Linux is an open-source operating system based on Unix. It is widely used because of its security, stability, and flexibility, and it supports multi-user and multitasking operations.

Software Requirements of Linux

Software requirements for Linux vary significantly depending on whether you are installing the core operating system or specific application suites. Generally, Linux has very minimal software dependencies because it is a self-contained family of open-source operating systems.

Software Requirements of Linux:

- A Linux distribution such as Ubuntu, Fedora, Debian, or CentOS is required to install and use the operating system.
- A boot loader like GRUB is necessary to load the Linux kernel during system startup.
- The Linux kernel is required to manage system resources such as CPU, memory, and hardware devices.
- A shell (for example, Bash) is needed to provide a command-line interface for user interaction.
- File system support such as EXT4 is required for storing and managing files securely.
- Device drivers are required to enable communication between the operating system and hardware components.

Hardware Requirements of Linux

Hardware requirements for Linux are flexible, ranging from "headless" servers that run on ancient chips to modern desktops requiring multi-core power.

- A 64-bit dual-core processor (2 GHz or higher) is required for modern desktop Linux distributions such as Ubuntu or Fedora.
- Minimum 4 GB RAM is required for desktop use, while 8 GB RAM is recommended for smooth multitasking.
- Linux requires at least 25 GB of free disk space, and an SSD is preferred for better performance.
- A graphics card with 3D acceleration and a minimum of 256 MB VRAM is required for graphical desktop environments.
- A display resolution of 1024×768 or higher is necessary for proper GUI operation.
- For server or headless Linux systems, a 1 GHz processor is sufficient.

Operating System 2: Unix

Introduction to Unix

Unix is one of the oldest operating systems, developed at AT&T Bell Labs, and is widely used in servers and enterprise systems due to its stability, multitasking, and multi-user support.

Software Requirements of Unix

Software Requirements of Unix include the Unix kernel, a command-line shell, standard system utilities, system libraries, and required runtimes for application execution.

- Unix Kernel is required as the core component to manage memory, processes, hardware devices, and file systems.
- A Shell is needed to act as an interface between the user and the kernel; common shells include sh, ksh, csh, and Bash.
- A hierarchical Unix file system is required to organize files and directories, where devices are also treated as files.
- Basic file management utilities such as ls, cp, mv, and rm are required for handling files and directories.
- Text processing tools like vi, grep, sed, and awk are essential for editing and manipulating text data.
- User management utilities such as passwd, who, and finger are required for user authentication and security.

Hardware Requirements of Unix

the physical resources needed to run the Unix operating system.

These include a compatible processor, sufficient RAM, adequate disk space, and supported hardware architecture.

- A minimum 1 GHz processor is required for most modern Unix systems, while multi-core processors are preferred for production environments.
- RAM of at least 1 GB is required for basic operation, and 4 GB or more is recommended for enterprise and multi-user workloads.
- Unix requires 5 GB to 10 GB of hard disk space for a basic installation.
- Swap space is essential for system stability and is typically configured as 2–3 times the physical RAM.
- A network interface card (NIC) supporting 10/100 Mbps or Gigabit Ethernet is required for communication and system management.
- IBM AIX requires hardware based on IBM PowerPC or RS/6000 processors.

Operating System 3: Windows XP

Introduction to Windows XP

Windows XP is a graphical operating system developed by Microsoft. It was widely used due to its simple interface and low hardware requirements. Windows XP supported multitasking and provided a stable environment for both home and office users. However, it is now obsolete and no longer supported by Microsoft.

Software Requirements of Windows XP

Software requirements for Windows XP revolve around its core architecture and essential runtime environments needed for applications to run effectively.

Software Requirements of Windows XP:

- Windows XP is built on the Windows NT kernel, which manages system resources and application execution.
- It supports NTFS, FAT32, and FAT16 file systems, with NTFS recommended for better security.
- Service Pack 3 (SP3) is required as the final and most stable update for 32-bit Windows XP systems.
- It includes DirectX 8.1, which can be upgraded to DirectX 9.0c for improved gaming and multimedia performance.
- Windows XP supports application runtimes such as .NET Framework up to version 4.0 and Visual C++ Redistributables (2005, 2008, 2010).
- It ships with Internet Explorer 6 (upgradable to Internet Explorer 8) and supports Windows Media Player 11 for internet and media functionality.

Hardware Requirements of Windows XP

Windows XP's hardware requirements were famously modest at launch, but they increased significantly as modern software and Service Packs (SP2 and SP3) were released.

Hardware Requirements of Windows XP :

- Processor: Requires a minimum Pentium 233 MHz processor; 300 MHz or faster is recommended for better performance.
- RAM: Minimum 64 MB RAM is required, but 128 MB or more is recommended; 512 MB–1 GB is ideal for smooth use with Service Pack 3.
- Storage: Needs at least 1.5 GB free hard disk space; 5 GB or more is recommended to support updates and applications.
- Display & Graphics: Requires Super VGA (800 × 600) resolution and a graphics card with at least 8 MB VRAM supporting DirectX 9.0c.

Operating System 4: Windows 7 / Windows 8

Introduction to Windows 7 and Windows 8

Windows 7 and Windows 8 are advanced operating systems developed by Microsoft with improved performance, security, and graphical interface. Windows 7 is known for its stability and user-friendly design, while Windows 8 introduced a touch-based interface and faster boot times.

These operating systems are widely used in personal computers, laptops, and professional environments.

Software Requirements of Windows 7 / 8

Software requirements for Windows 7 and Windows 8 are nearly identical, as both share the same underlying architecture.

Software Requirements of Windows 7 / 8

- Windows 7 and Windows 8 support both 32-bit (x86) and 64-bit (x64) system architectures and require the NTFS file system for the primary system partition.
- DirectX support is included by default, with DirectX 11 in Windows 7 and DirectX 11.1 in Windows 8 for enhanced graphics and multimedia performance.
- Internet Explorer 11 is supported on Windows 7 (via upgrade) and is included by default in Windows 8.1.
- Windows 7 requires manual installation of .NET Framework 4.5 or later, while Windows 8 includes .NET Framework 4.5 as a built-in component.
- Visual C++ Redistributable packages (versions 2010 to 2015–2022) are required for running many modern applications and games.
- Proper system operation requires WDDM 1.0 or higher graphics drivers, includes Windows Media Player 12, and depends on firmware support—Legacy BIOS for Windows 7 and UEFI with Secure Boot for Windows 8.

Hardware Requirements of Windows 7 / 8

Both Windows 7 and Windows 8 share the same baseline hardware requirements, as they were designed to run on the same generation of machines.

Hardware Requirements of Windows 7 / Windows 8 :

- Processor: 1 GHz or faster processor
- RAM: 1 GB for 32-bit version and 2 GB for 64-bit version
- Hard Disk Space: Minimum 16 GB for 32-bit and 20 GB for 64-bit
- Graphics Card: DirectX 9 compatible graphics device with WDDM driver
- Display: Minimum resolution of 1024×768
- Input Devices: Keyboard and mouse or compatible pointing device
- Other: Internet connection required for updates and activation